When Machine Learning Beats Persistence: A Leakage-Free Streamflow and Drought Benchmark for the Tropical Andes

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Abstract Machine-learning studies of Andean streamflow rarely compare their models with persistence, and their validation designs can leak information from the test period. We built an open benchmark of 19 long-record gauges from the Andean Glacierized Catchment dataset in three regions: the Altiplano, including Puente Isla Cabanillas in the Lake Titicaca basin, the Cordillera Blanca, and the western and southern Andes. Daily discharge was combined with ERA5 and MERRA-2 forcing and GloFAS v4 simulated discharge. Six learners were compared with persistence and climatology at 1-, 3- and 7-day leads under a calendar-continuous, embargoed protocol tuned on validation data only. Uncertainty was estimated with block-bootstrap intervals and Diebold–Mariano tests. The learners beat persistence at 15 of 19 gauges at one day and at 18 of 19 at seven days (median skill 0.101). Lower error did not improve drought detection: the median critical success index of drought days was 0.43 for persistence and 0.30 for the learners. Drought synchrony decayed over about 582 km, and skill depended on hydroclimatic regime rather than location.

Key words: Tropical Andes · Altiplano · Streamflow forecasting · Hydrological drought · Machine learning · Data leakage

1 Introduction

Rivers of the tropical Andes supply irrigation, hydropower and drinking water, but their flow is strongly seasonal and exposed to glacier retreat and drought [24]. On the Altiplano, rivers such as the Coata–Cabanillas feed Lake Titicaca and depend on a short austral-summer wet season [7], and hydrological droughts are projected to become longer and more frequent [24]. Short-range forecasts and early drought signals are therefore useful, but gauges are sparse and records are gappy.

Machine learning (ML) is widely used to forecast streamflow [14]. At daily leads, however, discharge is so autocorrelated that persistence, $\hat{Q}_{t+h} = Q_t$, is hard to beat [16]. Reported gains can be inflated by weak references, by shuffled cross-validation that leaks temporal information [2, 22], and by giving the baseline and the learners different information at the time of issue. An earlier version of our own pipeline made the last error: the learners saw discharge only up to $t-1$ while persistence used Q_t, and calendar gaps were treated as consecutive days. Correcting both reversed the one-day conclusion.

The aim of this study was to measure how much ML adds to persistence for 1- to 7-day streamflow and drought forecasts in the Andes when no information leaks from the test period. The paper makes four contributions: (i) an open benchmark of 19 gauges in three hydroclimatic regions; (ii) a leakage-free protocol checked by automated tests; (iii) skill against persistence with confidence intervals, compared with earlier Andean studies; and (iv) a direct evaluation of drought-day detection and of its spatial coherence. Section 2 reviews related work, Sect. 3 describes the data and methods, Sect. 4 presents the results, Sect. 5 discusses them with their limitations, and Sect. 6 concludes.

2 Related Work

Random forests and LSTM networks have been used for hourly runoff forecasting in Ecuadorian Andean catchments [3, 12, 13]. In Chile, LSTM networks supported flood warning in the Maipo basin [6], and random forests predicted monthly snowmelt-driven flows in the Elqui basin [15]. Support-vector regression was tested for monthly runoff in the Central Andes of Argentina [19]. In Peru, neural networks forecast monthly flows in the Crisnejas basin [21], and k-nearest neighbours forecast daily flows of the Ramis River on the Altiplano [11]. Table 1 summarizes these studies. Seven of the eight used a single target gauge, none compared the models with persistence, and none reported an uncertainty estimate for the skill. For Peruvian agricultural data, blocked cross-validation showed that random folds overstate accuracy [22]. Hydrological drought is a below-normal availability of streamflow [20], and its forecast needs an explicit identification step [18]. We therefore score drought days directly.

3 Data and Methods

Data. Figure 1 summarizes the workflow. We used quality-controlled (QC) daily discharge from the Andean-GC dataset [1]. We kept every Peruvian gauge of the National Meteorological and Hydrological Service (SENAMHI) and every Chilean Altiplano gauge of the General Directorate of Water (DGA) with at least 5000 QC days. This gave 19 gauges: 7 on the Altiplano, 6 in the glacier-fed Cordillera Blanca,

Table 1 Data-driven streamflow studies in the Andes. Pers.: persistence reference; CI: uncertainty of skill. Scores as reported by the authors.

Study	Region	Gauges	Step, lead	Method	Pers.	CI	Reported score
Muñoz et al. 2018 [13]	Ecuador	1	4–24 h	RF	no	no	NSE 0.76–0.38
Contreras et al. 2021 [3]	Ecuador	1	4–24 h	RF	no	no	R^2 0.69–0.29
Merizalde et al. 2023 [12]	Ecuador	1	1–11 h	LSTM	no	no	NSE 0.93–0.67
de la Fuente et al. 2019 [6]	Chile	9	hourly, 3 d	LSTM	no	no	NSE ≈0.93–0.99
Núñez et al. 2023 [15]	Chile	1	1–4 months	RF	no	no	R^2 0.85–0.73
Teverovsky et al. 2023 [19]	Argentina	1	1 month	SVR	no[a]	no	NSE up to 0.90
Vásquez-R. et al. 2023 [21]	Peru	1	monthly	ANN	no	no	r^2 0.81
Lujano et al. 2023 [11]	Peru (Altiplano)	1	1 day	k-NN	no	no	NSE 0.979
This study	Peru, Chile	19	1–7 days	6 learners	yes	yes	skill 0.101 at 7 days

[a]Compared with seasonal ARIMA.

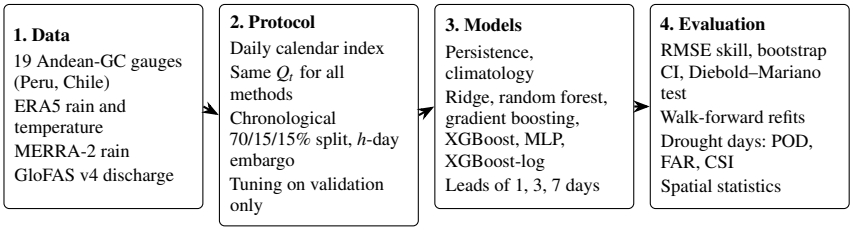


Fig. 1 Workflow of the benchmark. Code, tests and result tables are in the public repository.

and 6 in the western and southern Andes (Fig. 2). The focus gauge was Puente Isla Cabanillas (2843 km^2, mean elevation 4471 m), which drains into Lake Titicaca. Daily precipitation and 2-m temperature came from ERA5 [8] at each catchment centroid through Open-Meteo [23]. MERRA-2 precipitation and GloFAS v4 discharge were added in a data-fusion test.

Forecasts and learners. A forecast issued on day t predicts Q_{t+h}, $h \in \{1,3,7\}$ days, from 27 predictors available at t. These were discharge lags (0 to 30 days), 7- to 30-day rolling means and standard deviations, precipitation lags and 7- to 90-day totals, temperature means, and the day of year. A lag of k days is exactly k calendar days, and samples touching a gap were dropped. The references were persistence, using the same Q_t as the learners, and a day-of-year climatology. The six learners were Ridge regression, random forest, histogram gradient boosting, XGBoost, a two-layer perceptron (MLP), and an XGBoost trained on the change in $\log(1+Q)$ (XGBoost-log).

Protocol. Each record was split chronologically into 70% training, 15% validation and 15% test, with an h-day embargo between blocks. Hyper-parameters and the target form (level or change) were chosen on the validation root-mean-square error (RMSE). The chosen model was refit on training plus validation and scored once on test, so the test set never selected models. Ridge was also run in a three-fold expanding walk-forward at every gauge. Automated tests check calendar offsets, the absence of future information and the embargo.

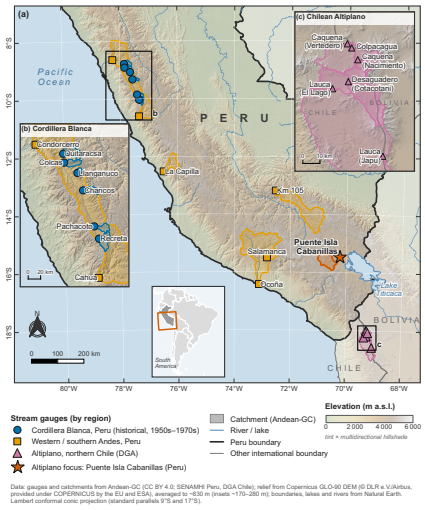


Fig. 2 The 19 gauges and their catchments by region. The focus gauge, Puente Isla Cabanillas, is highlighted.

Scores. We report the Nash–Sutcliffe efficiency (NSE) and the skill $S = 1 - \mathrm{RMSE}_{\mathrm{model}}/\mathrm{RMSE}_{\mathrm{pers}}$. Its 95% confidence intervals (CIs) came from a moving-block bootstrap [9], and equal accuracy was tested with the Diebold–Mariano (DM) test [4]. A day was in drought when Q fell below the monthly 20th percentile of the fitting period [20]. Detection was scored with the probability of detection (POD), false-alarm ratio (FAR) and critical success index (CSI) [17]. Drought synchrony was measured with the correlation of a nonparametric 30-day standardized streamflow index (SSI-30) [5] between gauge pairs, fitted as $\rho(d) = c\,e^{-d/d_0}$ against distance d. Spatial clustering of skill was tested with Moran's I.

4 Results

Focus gauge. At Cabanillas (Table 2; test period 2021-09-22 to 2024-12-24), the validation-selected random forest reached NSE = 0.922 at one day, against 0.902 for persistence. This is a skill of 0.110 (95% CI 0.064 to 0.144; DM p <0.001), the reverse of the result of our earlier, leaky pipeline. At seven days, persistence fell to NSE = 0.461. Ridge on the change target gained 0.115 with a CI above zero, and its walk-forward folds gave 0.166, 0.114, 0.114. The learner selected on validation at this lead (XGBoost-log) had a smaller, non-significant skill of 0.038.

All gauges. The learner beat persistence at 15/19, 15/19 and 18/19 gauges at 1, 3 and 7 days, and its CI excluded zero at 9, 10 and 12 gauges (Fig. 3). The median skill grew from 0.046 to 0.080 and 0.101, because persistence degraded faster than the learners. Between one and seven days it rose from 0.032 to 0.118 in the Cordillera

Table 2 Test scores at Puente Isla Cabanillas. Skill is relative to persistence, with 95% CI. †Learner selected on validation. CSI: drought-day detection at seven days.

	1-day lead		7-day lead		
Method	NSE	Skill [95% CI]	NSE	Skill [95% CI]	CSI
Persistence	0.902	–	0.461	–	0.61
Ridge	0.917	0.081 [0.03, 0.12]	0.578	0.115 [0.03, 0.21]	0.47
Random forest	0.922	0.110† [0.06, 0.14]	0.528	0.064 [−0.02, 0.14]	0.59
Hist. gradient boosting	0.916	0.074 [0.01, 0.14]	0.437	−0.022 [−0.16, 0.12]	0.18
XGBoost	0.915	0.071 [−0.01, 0.13]	0.407	−0.049 [−0.19, 0.09]	0.13
MLP	0.927	0.141 [0.09, 0.18]	0.203	−0.216 [−0.48, −0.01]	0.36
XGBoost-log	0.916	0.077 [0.00, 0.13]	0.502	0.038† [−0.08, 0.18]	0.56

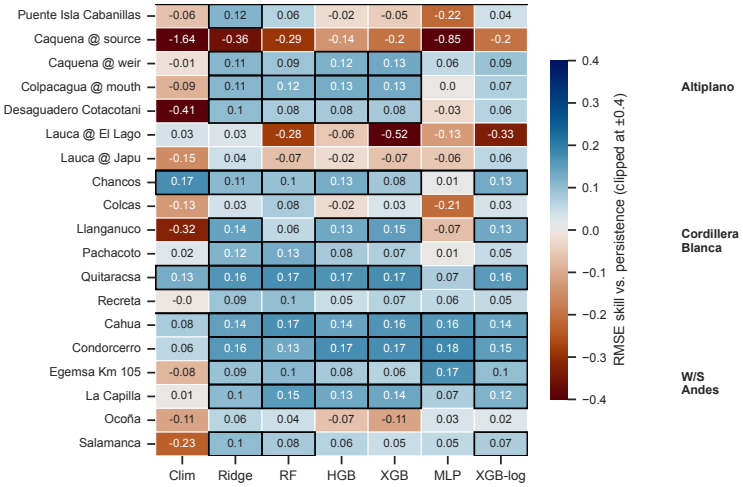


Fig. 3 Seven-day RMSE skill against persistence for every gauge and method. Boxed cells: 95% CI above zero. Gauges are grouped by region (top to bottom: Altiplano, Cordillera Blanca, western and southern Andes).

Blanca and from 0.047 to 0.110 in the western and southern Andes, but stayed flat on the Altiplano (0.081 to 0.057). Ridge alone was skilful at 18/19 gauges at seven days, and 53 of 57 walk-forward folds had positive skill (median 0.103). Adding MERRA-2 and GloFAS changed the seven-day Ridge skill by a median of only −0.004.

Drought detection. Lower RMSE did not give better detection of drought days (Fig. 4). At seven days, the median CSI was 0.43 for persistence and 0.30 for the learner, which beat persistence at only 2/19 gauges; XGBoost-log reached 0.33. The learners raised fewer false alarms (median FAR 0.35 vs. 0.39) but missed more drought days (POD 0.42 vs. 0.60).

Spatial statistics. The SSI-30 correlation of 104 gauge pairs decreased with distance (Spearman −0.56, $p < 0.001$), with $d_0 = 582$ km (95% CI 330–1069 km), and was higher within regions than across them (median 0.28 vs. 0.09). Skill showed no spatial clustering (Moran's $I = 0.018$, $p = 0.374$). It decreased with snow fraction

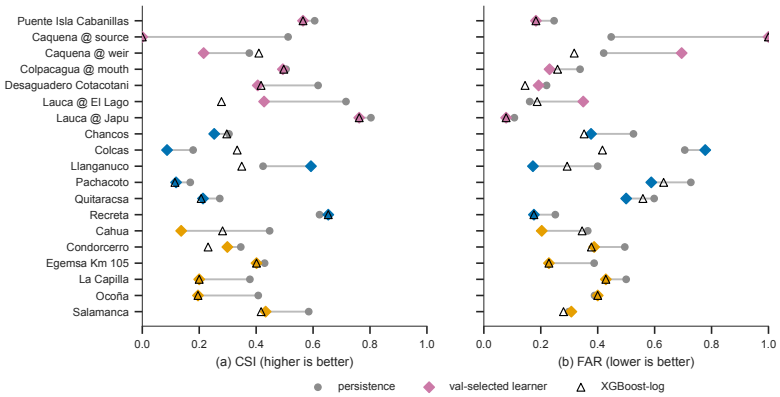


Fig. 4 Seven-day detection of drought days for each gauge. Left: CSI; right: false-alarm ratio.

(ρ =-0.57, Holm-adjusted $p = 0.071$), precipitation concentration and aridity, which describe the Altiplano headwaters.

5 Discussion

The largest change in our conclusions came from the protocol, not from the model. When persistence and the learners received the same information and lags followed the calendar, a one-day loss at Cabanillas became a significant gain. The gains are modest (median RMSE reduction of 5–10%) but consistent across gauges and folds, and linear models were as good as tree ensembles, as in large-sample studies [14].

This helps to read the earlier Andean studies (Table 1). Their scores are high, often NSE above 0.9 at short leads, but none was compared with persistence. At Cabanillas, persistence alone reached NSE $= 0.902$ at one day and the best learner 0.922. For the Ramis River, a k-NN model with only the previous day's discharge reached NSE $= 0.965$, against 0.979 for the best model [11]. In both rivers most of the NSE comes from persistence, so a high NSE is not evidence of skill at daily leads.

The Altiplano was the hardest region: its short convective pulses and long recessions are only partly resolved by a 0.25° reanalysis [7, 10]. Drought detection did not follow RMSE because models trained on squared error smooth the recession limb near the threshold, so drought warnings should be trained and verified on the drought decision itself [18].

Limitations. ERA5 is coarse for steep catchments. The Cordillera Blanca records predate MERRA-2 and GloFAS, so the fusion test covered 7 gauges only. One validation block can be unstable under non-stationarity, as the seven-day selection at Cabanillas showed. Predictors used observed forcing, not weather forecasts, and Bolivian records were not available.

6 Conclusion

We released an open, leakage-free benchmark for 1- to 7-day streamflow and drought forecasting at 19 Andean gauges. The learners beat persistence at 18/19 gauges at seven days (median skill 0.101) and at one day at Cabanillas, but the gains did not carry over to drought-day detection. Future work will add drought-aware losses, operational weather forecasts and more Altiplano gauges.

Declarations

Data and code availability. Discharge: Andean-GC, doi:10.5281/zenodo.18035801. Code, leakage tests, gauge list, result tables and supplementary figures: `https://github.com/Andre031222/streamflow-drought-altiplano`. **Author contributions.** R. A. Vilca-Solorzano conceived the study, wrote the software and the manuscript; R. Quispe-Vargas validated the results; D. M. Yana-Yucra supervised the study. All authors revised and approved the manuscript. **Competing interests.** None. **Use of AI.** Generative AI was used to improve the language; the authors reviewed the text and take full responsibility for it.

References

1. Aguayo, R., Zekollari, H., van Tiel, M., Bolibar, J., Van Tricht, L., Ayala Ramos, A.I., Ultee, L.: Andean Glacierized Catchment (Andean-GC) dataset. Zenodo dataset (2026). DOI 10.5281/zenodo.18035801
2. Bergmeir, C., Benítez, J.M.: On the use of cross-validation for time series predictor evaluation. Information Sciences **191**, 192–213 (2012). DOI 10.1016/j.ins.2011.12.028
3. Contreras, P., Orellana-Alvear, J., Muñoz, P., Bendix, J., Célleri, R.: Influence of random forest hyperparameterization on short-term runoff forecasting in an Andean mountain catchment. Atmosphere **12**(2), 238 (2021). DOI 10.3390/atmos12020238
4. Diebold, F.X., Mariano, R.S.: Comparing predictive accuracy. Journal of Business & Economic Statistics **13**(3), 253–263 (1995). DOI 10.1080/07350015.1995.10524599
5. Farahmand, A., AghaKouchak, A.: A generalized framework for deriving nonparametric standardized drought indicators. Advances in Water Resources **76**, 140–145 (2015). DOI 10.1016/j.advwatres.2014.11.012
6. de la Fuente, A., Meruane, V., Meruane, C.: Hydrological early warning system based on a deep learning runoff model coupled with a meteorological forecast. Water **11**(9), 1808 (2019). DOI 10.3390/w11091808
7. Garreaud, R., Vuille, M., Clement, A.C.: The climate of the Altiplano: observed current conditions and mechanisms of past changes. Palaeogeography, Palaeoclimatology, Palaeoecology **194**(1–3), 5–22 (2003). DOI 10.1016/S0031-0182(03)00269-4
8. Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L., Healy, S., Hogan, R.J., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C.,

Radnoti, G., de Rosnay, P., Rozum, I., Vamborg, F., Villaume, S., Thépaut, J.N.: The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society **146**(730), 1999–2049 (2020). DOI 10.1002/qj.3803

9. Künsch, H.R.: The jackknife and the bootstrap for general stationary observations. The Annals of Statistics **17**(3), 1217–1241 (1989). DOI 10.1214/aos/1176347265
10. Lujano, E., Diaz, R.D., Lujano, R., Sanchez-Delgado, M., Lujano, A.: Hydrological performance of gridded meteorological products in Peruvian Altiplano basins. RBRH – Revista Brasileira de Recursos Hídricos **30**, e10 (2025). DOI 10.1590/2318-0331.302520240068
11. Lujano, E., Lujano, R., Huamani, J.C., Lujano, A.: Hydrological modeling based on the KNN algorithm: An application for the forecast of daily flows of the Ramis river, Peru. Tecnología y Ciencias del Agua **14**(2), 169–203 (2023). DOI 10.24850/j-tyca-14-02-05
12. Merizalde, M.J., Muñoz, P., Corzo, G., Muñoz, D.F., Samaniego, E., Célleri, R.: Integrating geographic data and the SCS-CN method with LSTM networks for enhanced runoff forecasting in a complex mountain basin. Frontiers in Water **5**, 1233,899 (2023). DOI 10.3389/frwa.2023.1233899
13. Muñoz, P., Orellana-Alvear, J., Willems, P., Célleri, R.: Flash-flood forecasting in an Andean mountain catchment—development of a step-wise methodology based on the random forest algorithm. Water **10**(11), 1519 (2018). DOI 10.3390/w10111519
14. Nearing, G.S., Kratzert, F., Sampson, A.K., Pelissier, C.S., Klotz, D., Frame, J.M., Prieto, C., Gupta, H.V.: What role does hydrological science play in the age of machine learning? Water Resources Research **57**(3), e2020WR028,091 (2021). DOI 10.1029/2020WR028091
15. Núñez, J., Cortés, C.B., Yáñez, M.A.: Explainable artificial intelligence in hydrology: Interpreting black-box snowmelt-driven streamflow predictions in an arid Andean basin of north-central Chile. Water **15**(19), 3369 (2023). DOI 10.3390/w15193369
16. Pappenberger, F., Ramos, M.H., Cloke, H.L., Wetterhall, F., Alfieri, L., Bogner, K., Mueller, A., Salamon, P.: How do I know if my forecasts are better? Using benchmarks in hydrological ensemble prediction. Journal of Hydrology **522**, 697–713 (2015). DOI 10.1016/j.jhydrol.2015.01.024
17. Schaefer, J.T.: The critical success index as an indicator of warning skill. Weather and Forecasting **5**(4), 570–575 (1990). DOI 10.1175/1520-0434(1990)005<0570:TCSIAA>2.0.CO;2
18. Sutanto, S.J., Van Lanen, H.A.J.: Streamflow drought: implication of drought definitions and its application for drought forecasting. Hydrology and Earth System Sciences **25**(7), 3991–4023 (2021). DOI 10.5194/hess-25-3991-2021
19. Teverovsky Korsic, S.A., Notarnicola, C., Uriburu Quirno, M., Cara, L.: Assessing a data-driven approach for monthly runoff prediction in a mountain basin of the Central Andes of Argentina. Environmental Challenges **10**, 100,680 (2023). DOI 10.1016/j.envc.2023.100680
20. Van Loon, A.F.: Hydrological drought explained. WIREs Water **2**(4), 359–392 (2015). DOI 10.1002/wat2.1085
21. Vásquez-Ramírez, L., Vásquez-Paredes, L.F.: Predicción de caudales mensuales en ríos de cuencas altoandinas con enfoque de redes neuronales artificiales. Caso: río Crisnejas, Perú. Tecnología y Ciencias del Agua **14**(1), 124–199 (2023). DOI 10.24850/j-tyca-14-01-04
22. Vilca Apaza, W., Torres Cruz, F., Carpio Vargas, E.E.: Blocked and hierarchical cross-validation for agricultural data with spatial autocorrelation: Application to Peru's national agricultural survey. In: S. Duarte Correia, M. Lahby (eds.) Green Technologies and Sustainable Intelligence—Volume II: Applications, Case Studies, and Future Challenges, *Springer Proceedings in Physics*, vol. 352, pp. 3–13. Springer Nature Switzerland, Cham (2026). DOI 10.1007/978-3-032-20151-5_1
23. Zippenfenig, P.: Open-Meteo.com Weather API. Zenodo software, version 1.4.0 (2024). DOI 10.5281/zenodo.7970649
24. Zubieta, R., Molina-Carpio, J., Laqui, W., Sulca, J., Ilbay, M.: Comparative analysis of climate change impacts on meteorological, hydrological, and agricultural droughts in the Lake Titicaca basin. Water **13**(2), 175 (2021). DOI 10.3390/w13020175